



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2024

BIOLOGY (N530)

TIME: 45 MINUTES

Candidates answer on the Question Paper.
Additional Materials: Ruler, Calculator

READ THESE INSTRUCTIONS FIRST

- 1. Write your index number in the space provided above.
- 2. Write in dark blue or black ink. Do not use correction fluid.
- 3. You may use a soft pencil for any diagram, graph or rough working.
- 4. Diagrams are not drawn to scale unless otherwise specified.
- 5. Any rough working should be done in this booklet.
- 6. Answer **ALL** questions.
- 7. This document consists of **5** questions printed on **10** pages, numbered **2** to **11**.
- 8. The number of marks is given at the end of each question or part question.
- 9. The total number of marks for this paper is **50**.

For Examiners' use							
Question No.	1	2	3	4	5	Total	Signature
Examiner							
Team Leader							
CE/ACE							

Question 1 (10 marks)

Circle the correct answer. Each item carries one mark.

1. Which component of blood helps in blood clotting?

- A** Red blood cells
- B** White blood cells
- C** Plasma
- D** Platelets

2. **Fig. 1.1** shows a section through an artery.

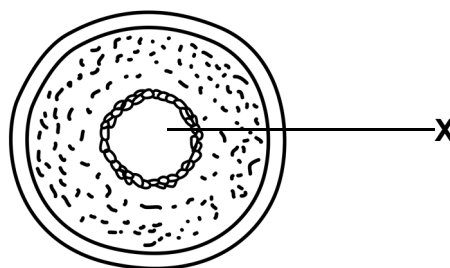


Fig. 1.1 : Section through an artery

What does **X** in **Fig. 1.1** represent?

- A** A small lumen
- B** A large lumen
- C** A thick layer of muscle and elastic fibres
- D** A thin layer of muscle and elastic fibres

3. **Fig. 1.2** represents the process of asexual reproduction.

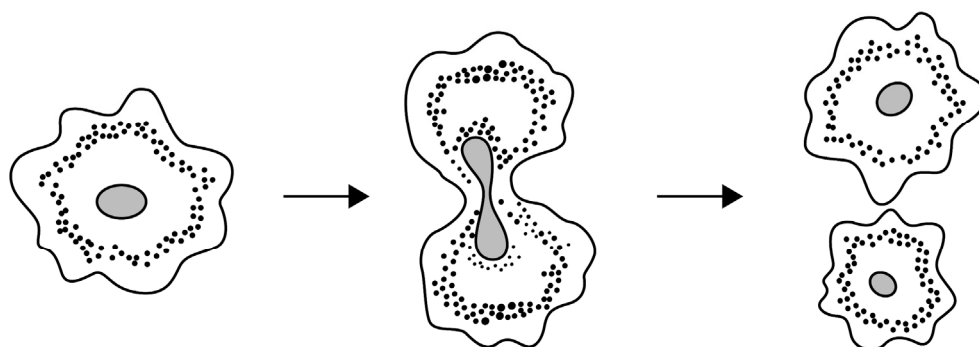


Fig. 1.2 : Asexual reproduction

Which one of the following will reproduce asexually?

- A** Amoeba
 - B** Cats
 - C** Elephants
 - D** Pigeons
4. Which of the following is a sexually transmitted disease?
- A** Cholera
 - B** COVID-19
 - C** Gonorrhoea
 - D** Stroke
5. Which of the following may lead to an increase in the spread of AIDS?
- A** Eating healthy food
 - B** Doing regular exercise
 - C** Using condoms during sexual intercourse
 - D** Sharing contaminated needles among drug users

6. **Fig. 1.3** shows the front view of the female reproductive system.

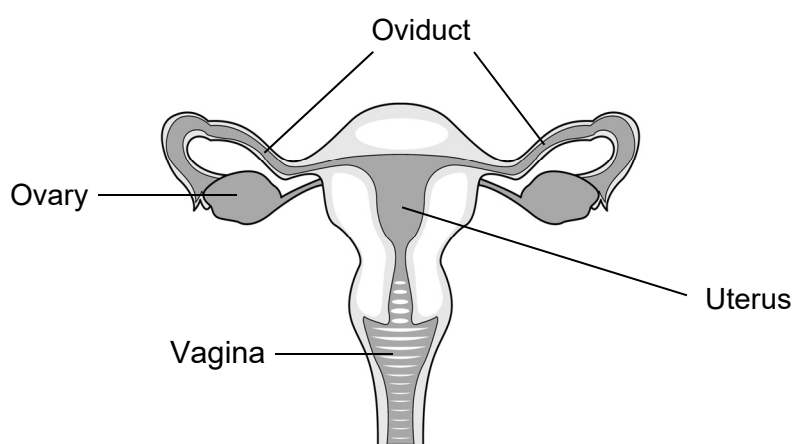


Fig. 1.3 : Front view of the female reproductive system

Which part of the female reproductive system produces eggs?

- A** Ovary
 - B** Oviduct
 - C** Uterus
 - D** Vagina
7. Which of the following can be estimated using a quadrat?
- A** Algae
 - B** Birds
 - C** Dogs
 - D** Fish
8. Which green pigment, present in leaves, traps light energy?
- A** Haemoglobin
 - B** Chlorophyll
 - C** Plasma
 - D** Starch

9. **Fig. 1.4** shows the structure of a leaf.

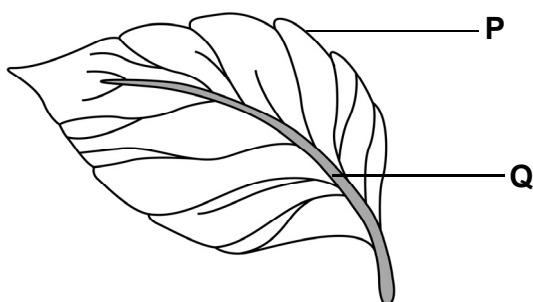


Fig. 1.4 : Structure of a leaf

What do **P** and **Q** represent?

	P	Q
A	network of veins	midrib
B	leaf apex	network of veins
C	leaf margin	network of veins
D	leaf margin	midrib

10. Why should a leaf be heated in alcohol before testing for starch?

- A** To stop all chemical reactions in the leaf
- B** To make the leaf softer
- C** To remove colour from the leaf
- D** To remove oxygen from the leaf

E

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Question 2 (7 marks)

(a) **Fig. 2.1** shows features of sexual reproduction in humans.

Using the words provided below, identify **A**, **B**, **C** and **D**.

zygote	foetus	ovum	sperm	embryo
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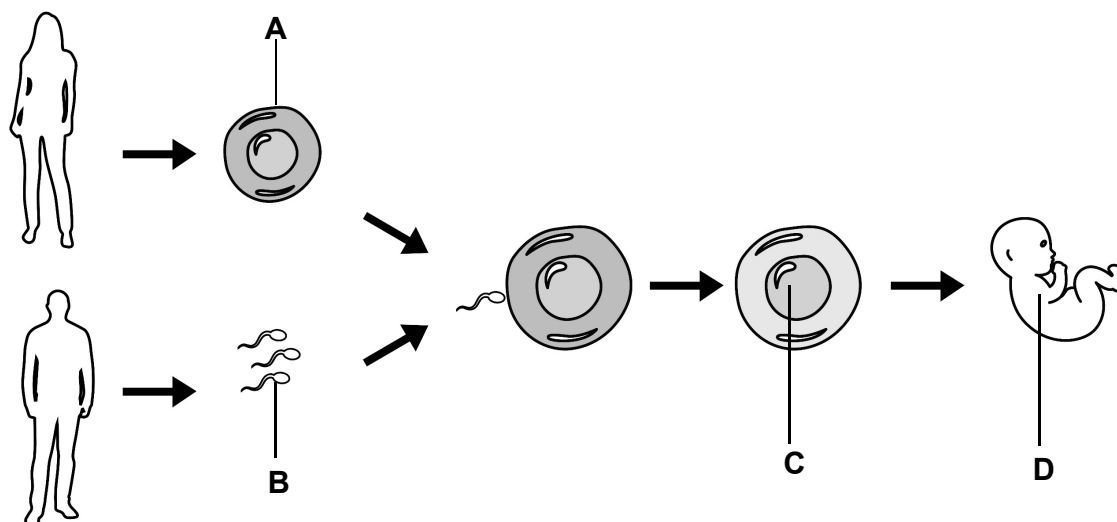


Fig. 2.1 : Sexual reproduction in humans

A :

B :

C :

D :

[4]

(b) Some plants reproduce asexually.

(i) Give one difference between sexual and asexual reproduction.

.....

[1]

(ii) Give two examples of plants that reproduce asexually.

1.
2.

[2]

E
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Question 3 (10 marks)

- (a) (i) Define biodiversity.

.....

.....

[2]

- (ii) Biodiversity is affected by natural calamities and human activities.

Complete the table below to indicate whether the following words are examples of **natural calamities** or **human activities**.

cyclone	pollution	deforestation	drought
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Natural calamities	Human activities

[4]

- (b) Give two **ecological** benefits of biodiversity.

1.
2.

[2]

- (c) Give two **socio-economic** benefits of biodiversity.

1.
2.

[2]

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Question 4 (11 marks)

(a) **Fig. 4.1** shows part of the circulatory system in humans.

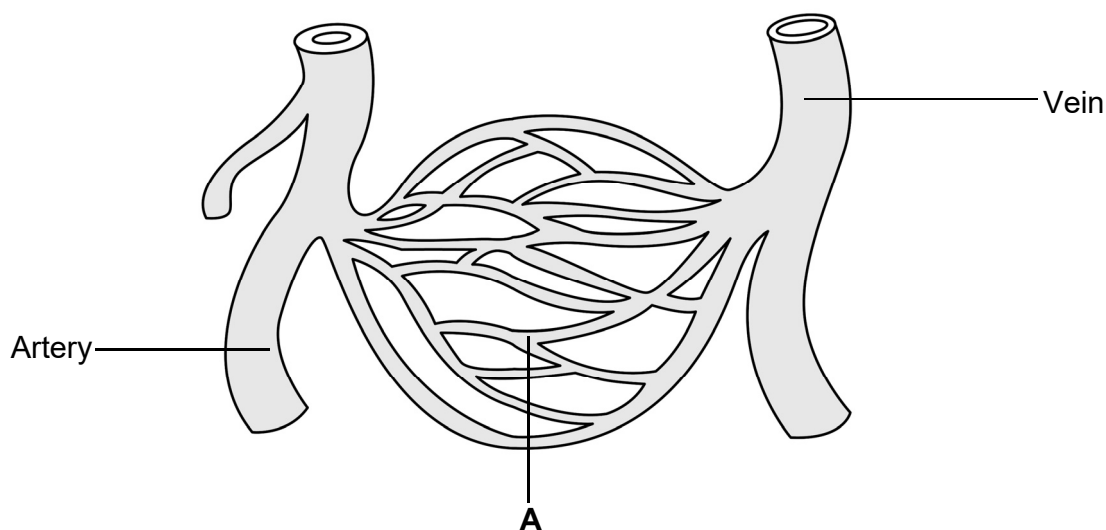


Fig. 4.1 : Part of the circulatory system

- (i) What does structure **A** represent?
 [1]
- (ii) Give the function of structure **A**.
 [1]
- (b) Give two differences in the way arteries and veins carry blood.
1.
 [2]
2.
 [2]
- (c) How is the stretching and recoiling of the artery walls known as?
 [1]
- (d) State two factors that may increase the risk of coronary heart disease.
1.
 2. [2]

(e) White blood cells are important components of blood.

(i) Describe how white blood cells protect the body from germs.

.....

.....

[2]

(ii) **Fig. 4.2** shows a white blood cell with a magnification of x 8000.

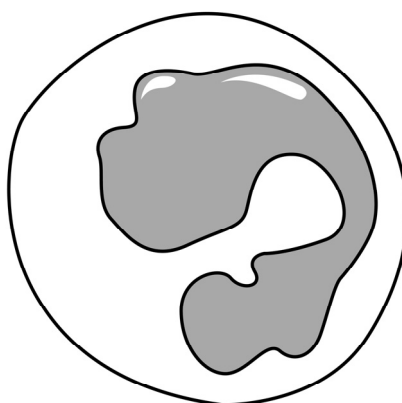


Fig. 4.2 : Magnified white blood cell

Calculate the actual size of the white blood cell in Fig. 4.2.
Show your working.

Answer mm

[2]

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Question 5 (12 marks)

(a) Leaves contain small pores.

(i) How are the small pores in the leaves called?

.....

[1]

(ii) What is the function of the small pores?

.....

[1]

(b) A student investigates the effects of light and carbon dioxide on photosynthesis.

Fig. 5.1 shows the set-up used for the investigation. The experiment is carried out in a dark room. The only source of light is a lamp.

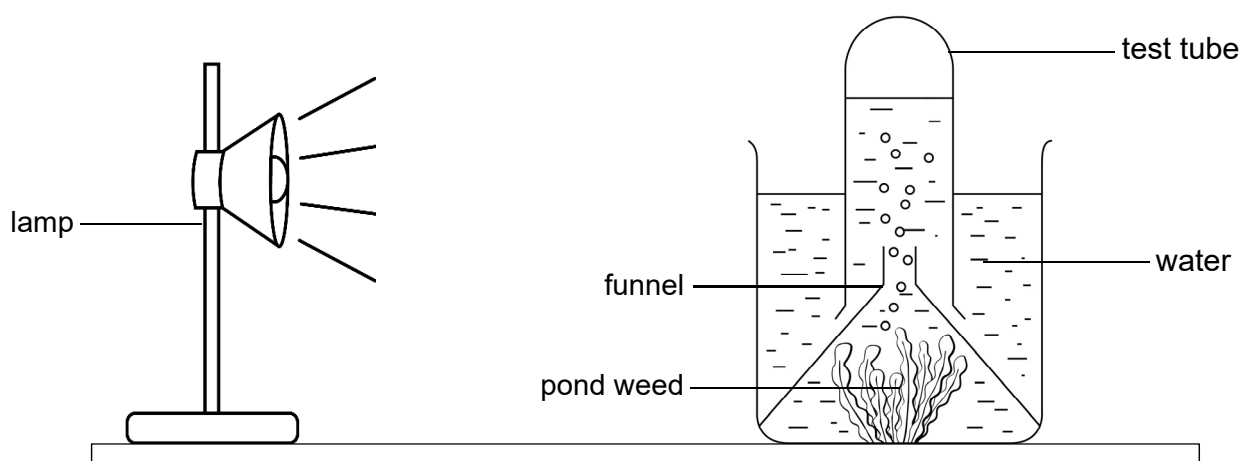


Fig. 5.1: Experiment on photosynthesis

The student carries out the experiment four times. He records his findings in the table below each time.

Test	Conditions		No of bubbles of gas produced/minute
	Light	Carbon dioxide source	
1	Present	Absent	0
2	Absent	Present	0
3	Present	Present	20
4	Absent	Absent	0

- (i) Give one variable that needs to be kept constant in each experiment.

.....

[1]

- (ii) Give two conclusions that the student can draw from this investigation.

.....

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.....

.....

[2]

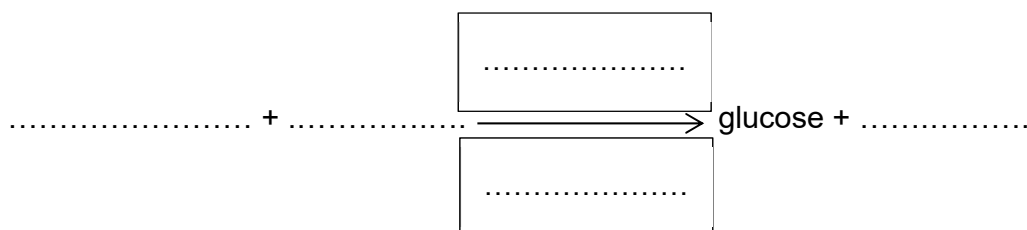
- (iii) Give one way how the set-up can be modified to increase the rate of photosynthesis.

.....

.....

[1]

- (c) (i) Complete the word equation to represent the process of photosynthesis.



[5]

- (ii) What happens to the excess glucose during photosynthesis?

.....

[1]

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